

# Computer Vision-Assisted Semantic Material Identification for High-Fidelity Digital Twins

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## 고정밀 디지털 트윈 구축을 위한 컴퓨터 비전 기반 시맨틱 재질 식별

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### Abstract

Ray-tracing-based digital twins critically depend on accurate electromagnetic material modeling, yet reconstructed meshes are often imported with simplified or default material assignments, degrading radio propagation predictions. This paper proposes a computer vision-assisted semantic material identification pipeline that converts a reconstructed 3D mesh into a high-fidelity scene for Sionna RT. Simulation results show that the proposed pipeline yields measurable, position-dependent channel differences, supporting 5G/6G digital twin construction.

### I. Introduction

Wireless digital twins enable site-specific 5G/6G network optimization through radio frequency (RF) propagation simulation in virtual replicas of physical environment. RF propagation can be modelled using ray-tracing tools, e.g. Sionna RT [1], based on the geometric and electromagnetic (EM) properties of the environment. Consequently, ray-tracing accuracy strongly depends on mesh material modeling, particularly at higher frequencies [2]. However, reconstructed meshes are typically simplified to a single homogeneous material (e.g., concrete) to reduce manual effort. To automate this process, computer vision (CV)-based semantic material classification has been proposed in [3], although material modeling remains simplified and demonstrated use cases are limited. This paper proposes a CV-assisted material identification pipeline that combines SAM 2.1 [4] and DINOv2 [5] with five hand-labeled UAV photographs to build a material-aware Sionna RT scene of the FTC building at Hanyang University, evaluated against a single-concrete baseline at 28 GHz.

### II. Proposed Pipeline

As shown in Fig. 1, the proposed pipeline consists of four stages. First, the scene is reconstructed by combining terrain and surrounding buildings imported into Blender through the Blomsm plug-in with the FTC building mesh generated via UAV photogrammetry using a DJI Mini 5 Pro. Next, five UAV photographs are manually labeled with concrete, glass, and metal seeds, yielding 152 labeled segments. A logistic regression classifier is then trained over frozen DINOv2 embeddings extracted from these segments, achieving a 5-fold balanced segment-level accuracy of 0.848. Finally, SAM 2.1 generates class-agnostic masks for the remaining UAV images, and each mask is classified using the trained logistic regression head.

The predicted labels are back-projected to mesh triangles via per-triangle majority voting, and a material-aware Sionna RT XML scene is exported by mapping concrete walls/roofs to ITU concrete, windows to glass, frames to metal, terrain mesh to a 28 GHz constant ground material ( $\epsilon_r=3.0$ ,  $\sigma=0.05$  S/m) since the ITU very\_dry\_ground model is defined up to 10 GHz, and unclassified surfaces to concrete.

### III. Channel-Level Evaluation Framework

We denote the channel frequency response (CFR) of the material-aware scene as  $h_{mat}$  and the CFR of the single-concrete

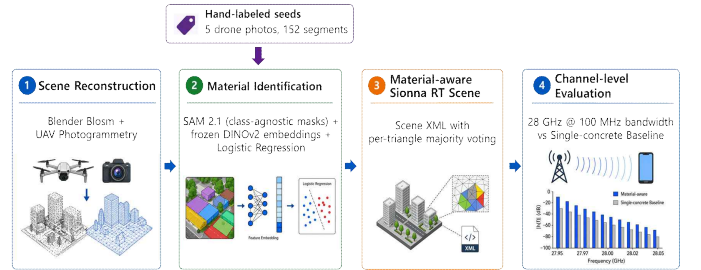


Fig. 1. Proposed CV-assisted material identification pipeline

baseline as  $h_{base}$ , extracted along an identical UE trajectory.

The overall power discrepancy between the two CFR vectors is measured by the normalized mean squared error (NMSE) over all subcarriers, defined in linear scale as

$$NMSE = \frac{\sum |h_{mat} - h_{base}|^2}{\sum |h_{mat}|^2}. \quad (1)$$

The NMSE in dB scale,  $10\log_{10}(NMSE)$ , approaches  $-\infty$  dB for identical CFRs and becomes 0 dB when the error energy equals the material-aware CFR energy.

The cosine similarity  $\rho$  captures the structural alignment between the two complex CFR vectors irrespective of amplitude, given by

$$\rho = \frac{|h_{mat}^H h_{base}|}{\|h_{mat}\| \cdot \|h_{base}\|}, \quad (2)$$

where  $(\cdot)^H$  denotes the Hermitian transpose. Values near 1 indicate near-parallel CFRs, lower values indicate differing spectral shapes.

Per-position received-signal-strength (RSS) error is reported in signed form to preserve the direction of mismatch.

$$\Delta P_{RSS} = 10\log_{10}|h_{mat}|^2 - 10\log_{10}|h_{base}|^2 \text{ [dB]}. \quad (3)$$

A positive  $\Delta P_{RSS}$  indicates that the single-concrete baseline underestimates received power compared to the material-aware scene.

Denoting the instantaneous per-subcarrier signal-to-noise ratio (SNR) as  $\gamma(f) = P_{tx}|h(f)|^2/N$ , a Shannon-style throughput proxy  $T(h)$  summarizes the link-level channel quality as

$$T(h) = B \cdot E_f[\log_2(1 + \gamma(f))], \quad (4)$$

where  $E_f[\cdot]$  denotes the subcarrier grid expectation,  $B = 100$  MHz is the bandwidth,  $P_{tx} = 30$  dBm is the transmit power, and the noise

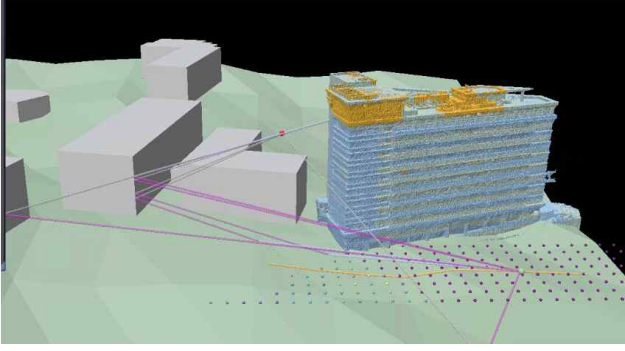


Fig. 2. 28 GHz propagation analysis with the Sionna RT GUI

power  $N = -174 + 10\log_{10}(B) + NF = -87$  dBm with noise figure  $NF = 7$  dB. In (4),  $P_{tx}$  and  $N$  are converted to linear power units before computing  $\gamma(f)$ . Equation (4) is a Shannon upper bound under the stated assumptions and is not a calibrated 5G NR throughput. In addition to these scalar metrics, per-position RSS curves provide spatial views of where material mismatch most affects the channel.

#### IV. Experimental Setup and Results

Ray-tracing simulations are performed at a carrier frequency of 28 GHz with 100 MHz bandwidth, one TX array, a UE trajectory of approximately 50 m resampled to 50 positions, and a maximum ray-tracing depth of 6. Fig. 2 shows the Sionna RT GUI with TX/RX, trajectory, and ray-path overlays.

Table 1 reports the canonical seed=42 metrics together with seed-averaged statistics across seeds 42-45. The global NMSE between the two scenes is  $-11.89$  dB for the canonical run and  $-25.0 \pm 9.8$  dB across seeds 42-45. In the canonical run, 28 of 50 UE positions exhibit material-sensitive channel responses (per-position NMSE  $> -60$  dB; seed-averaged  $24.3 \pm 2.6$  across seeds 42-45), while late-trajectory positions approach a numerical floor where the two scenes are nearly indistinguishable.

**Table 1. Material-aware vs single-concrete comparison (28 GHz, canonical seed = 42)**

| Metric                         | Material-aware | Single-concrete | $\Delta$ |
|--------------------------------|----------------|-----------------|----------|
| Mean RSS, all (dB)             | -139.94        | -140.50         | +0.56    |
| Mean RSS, m.s. (dB)            | -129.36        | -130.36         | +1.00    |
| Mean $T_{proxy}$ (Mbps)        | 89.22          | 99.49           | -10.27   |
| Mean $T_{proxy}$ , m.s. (Mbps) | 128.53         | 146.87          | -18.34   |
| Mean path count                | 12.4           | 12.8            | -0.4     |

\* m.s. = material-sensitive subset (per-position NMSE  $> -60$  dB,  $n=28/50$ ).  $\Delta$  = material-aware - single-concrete.

Although the average active path count differs by only 0.4 (12.4 vs 12.8), the throughput-proxy gap of 18 Mbps in the material-sensitive subset suggests that path amplitude and phase, rather than number of paths, drive the divergence. Fig. 3 visualizes this RSS comparison, where the two scenes diverge by up to 22 dB at a single outlier position ( $p=8$ ) and several dB in the material-sensitive region.

Fair-condition comparison (Table 2) shows similar active global NMSE at 3.5 GHz and 28 GHz, indicating that material-mapping sensitivity is position-dependent rather than frequency-dependent on the considered trajectory.

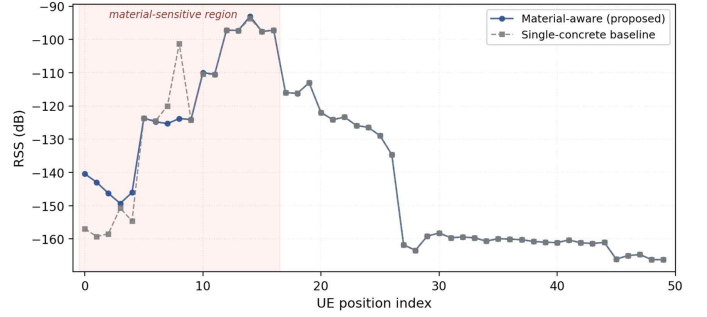


Fig. 3. Per-position RSS comparison at 28 GHz

**Table 2. Fair-condition frequency comparison (1M samples, max depth 6)**

| Metric                        | 3.5 GHz | 28 GHz  |
|-------------------------------|---------|---------|
| Active Global NMSE (dB)       | -19.53  | -19.60  |
| Active mean cos. sim.         | 0.444   | 0.430   |
| Mean $T_{proxy}$ error (Mbps) | 74.9    | 13.8    |
| both active / base only       | 32 / 18 | 32 / 18 |

\* both active / base only: number of UE positions where both scenes (or only baseline) yield active channel.

#### V. Conclusion

This paper proposes a CV-assisted material identification pipeline integrating SAM 2.1, DINOv2, and a logistic-regression head trained on five hand-labeled UAV photographs. Combined with a CFR-based evaluation framework, the pipeline produces position-dependent material-aware channel responses at 28 GHz with measurable differences from a single-concrete baseline, supporting site-specific 5G/6G digital twin construction. Future work includes hand-label expansion, integration with texture-aware platforms beyond Sionna RT, and exploration of alternative UE trajectories and TX/RX placements to enlarge the material-sensitive subset.

#### ACKNOWLEDGMENT

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